

Hypothesis Testing – t-test & Chi-Square

1 Concept Explanation (Simple & Clear)

Term	Meaning
Hypothesis	A claim or assumption we want to test.
Null Hypothesis (H_0)	No difference or no effect (status quo).
Alternative Hypothesis (H_1)	There is a difference or effect.
p-value	Probability of results happening by chance.
Significance Level (α)	Threshold to reject H_0 (commonly 0.05).
 Goal of Hypothesis Testing:	
To decide if a result is statistically significant or just happened by chance.	

2 Types of Tests Covered

Test	Used For	Example
t-test	Compare means of two groups	Avg. height of boys vs girls
Chi-Square Test	Compare categories / frequencies	Does gender affect buying choice?

3 Practical Example – Real World

Example 1: t-test (Mean Comparison)

A health researcher tests if a new diet changes average weight.

- H_0 : No difference in weight after diet
- H_1 : Diet causes weight change
If p-value < 0.05 → Diet works 

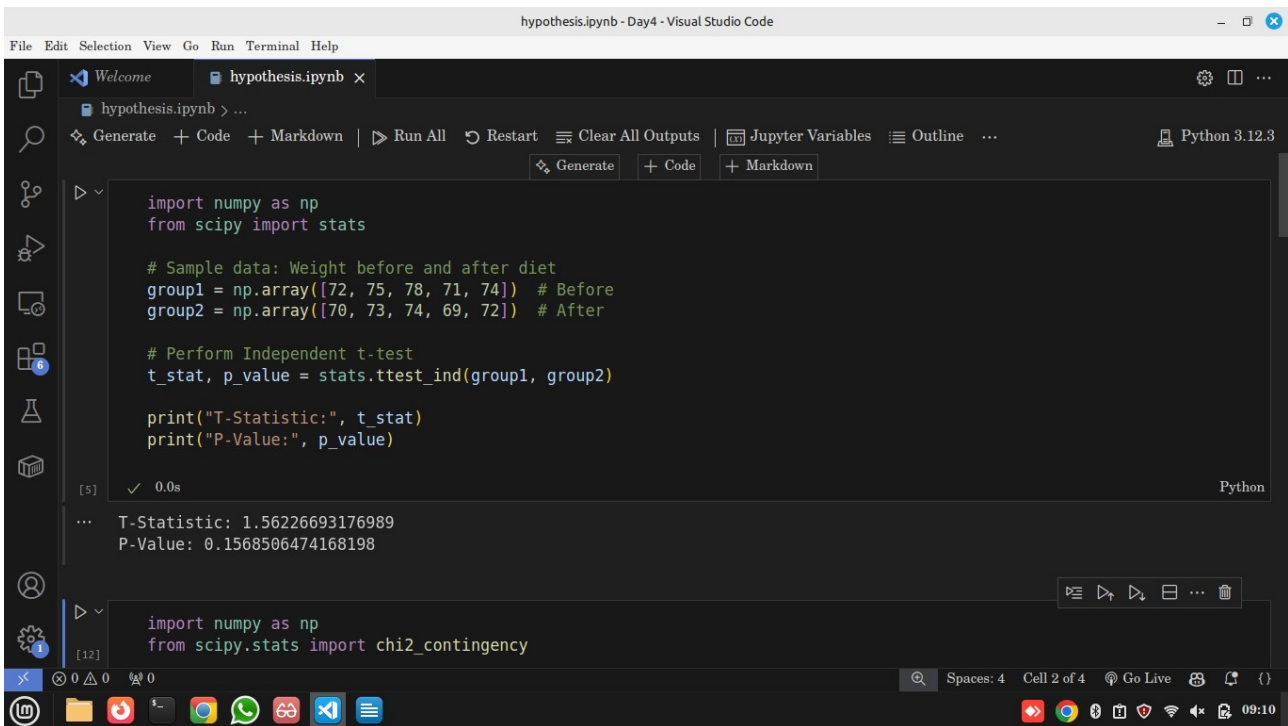
Example 2: Chi-Square (Category Relationship)

A store checks if gender affects product preference (Shoes vs Clothes).

- H_0 : Gender and choice are independent
 - H_1 : Gender influences choice
-

4 Python Code (With Explanation)

A) t-test in Python



```
import numpy as np
from scipy import stats

# Sample data: Weight before and after diet
group1 = np.array([72, 75, 78, 71, 74]) # Before
group2 = np.array([70, 73, 74, 69, 72]) # After

# Perform Independent t-test
t_stat, p_value = stats.ttest_ind(group1, group2)

print("T-Statistic:", t_stat)
print("P-Value:", p_value)
```

[5] ✓ 0.0s

... T-Statistic: 1.56226693176989
P-Value: 0.1568506474168198

```
import numpy as np
from scipy.stats import chi2_contingency
```

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Code Explanation:

- `ttest_ind()` → Compares means of two groups.
- `p_value < 0.05` → Reject H_0 → Significant difference.

B) Chi-Square Test in Python

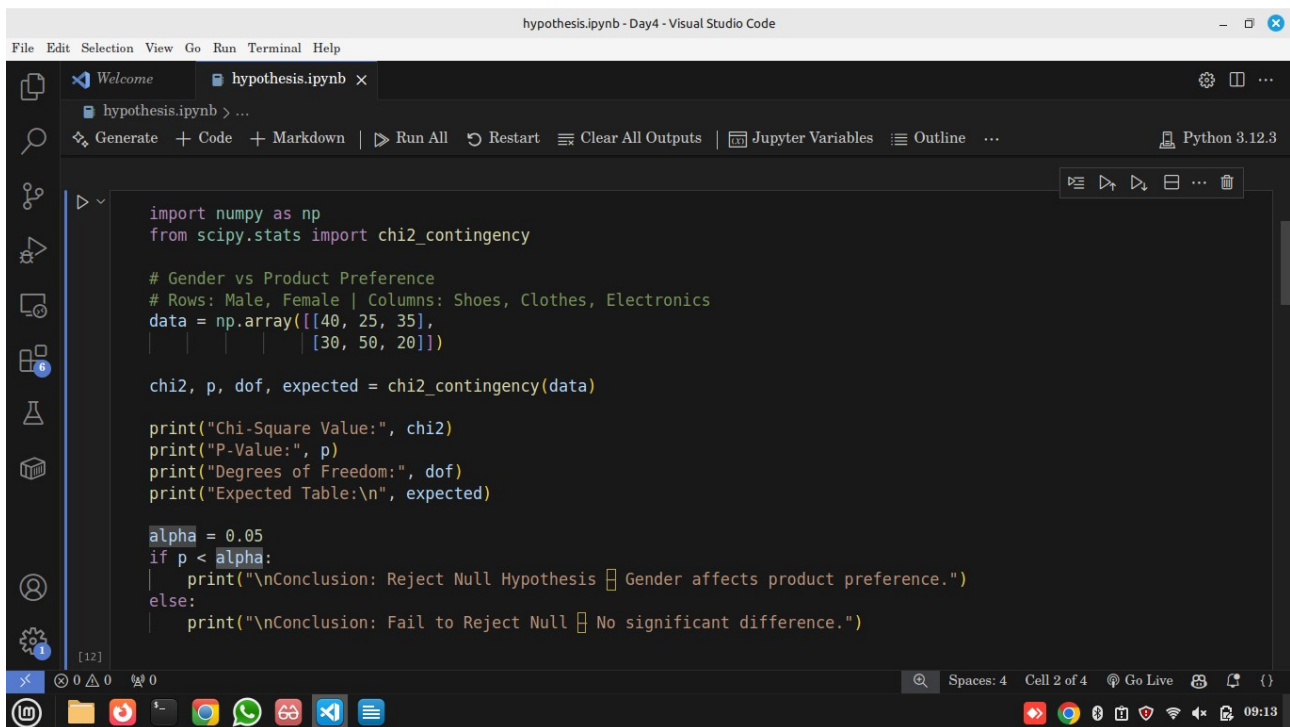
Example: Gender vs Product Preference

Gender	Shoes	Clothes	Electronics
Male	40	25	35
Female	30	50	20

Hypotheses for Chi-Square Test

Type of Hypothesis	Statement
Null Hypothesis (H_0)	Gender and product preference are independent. That means males and females choose products in the same pattern — gender does not affect what they prefer.
Alternative Hypothesis (H_1)	Gender and product preference are dependent. That means males and females have different product preferences — gender influences what they prefer.

✓ Python Code



```
import numpy as np
from scipy.stats import chi2_contingency

# Gender vs Product Preference
# Rows: Male, Female | Columns: Shoes, Clothes, Electronics
data = np.array([[40, 25, 35],
                 [30, 50, 20]])

chi2, p, dof, expected = chi2_contingency(data)

print("Chi-Square Value:", chi2)
print("P-Value:", p)
print("Degrees of Freedom:", dof)
print("Expected Table:\n", expected)

alpha = 0.05
if p < alpha:
    print("\nConclusion: Reject Null Hypothesis ☐ Gender affects product preference.")
else:
    print("\nConclusion: Fail to Reject Null ☐ No significant difference.")
```

📖 Code Explanation (in Simple English)

Line	Explanation
<code>data = np.array(...)</code>	This is our observed data from a survey.
<code>chi2_contingency(data)</code>	Performs Chi-Square Test .
<code>chi2</code>	Measures difference between observed vs expected.
<code>p</code>	Probability — if p < 0.05 , gender affects preference.
<code>expected</code>	What counts should look like if no relationship exists.

💡 Insight for Students

- If **p < 0.05** → Gender influences what people buy 🛍️
- If **p > 0.05** → No difference → Men & women buy randomly

💡 5 Insights – Why It Matters

Insight	Importance
Supports Data-Driven Decisions	No guessing, pure statistics
Used in A/B Testing	Marketing, design experiments
Validates Scientific Research	Medicine, psychology, business

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Real-World Applications

Field	Real Use Case	How Hypothesis Testing is Used
Health & Medicine	Test drug effectiveness	Researchers compare two groups: one using the new drug and one using a placebo. They perform a t-test to check if the drug truly lowers blood pressure more than placebo. If $p < 0.05$, they conclude the drug works.
Business & Tech	A/B Test for Website Design or Ads	A company shows Design A to 100 users and Design B to another 100 users. They compare average clicks or sales using a t-test to see which design performs better. Helps make data-driven decisions , not guesses.
Education	Compare teaching methods	A school tests if a new teaching method improves exam scores. Group 1 learns traditionally, Group 2 uses new method. A t-test checks if score differences are significant or random.
Marketing	Gender purchase behavior (Chi-Square Test)	A store checks if males and females buy different types of products (shoes, clothes, electronics). A Chi-Square Test is used to see if buying choice depends on gender OR if it's random.
Manufacturing	Quality Control	Factories test if products from Machine A and Machine B have the same defect rate. Hypothesis testing helps decide if a machine needs repair.
Sports Science	Athlete Performance Comparison	Coaches use t-tests to compare average running times before and after training programs to see if training truly improved performance.
Psychology	Behavior & Treatment Studies	Psychologists test if therapy changes patient anxiety scores. Hypothesis testing confirms if improvement is real, not by luck.

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15-Minute Assignment (Class Activity)

Task: Perform a t-test

- Collect test scores of **two classes** (Class A & B, 5 students each)
- Use Python to test if average scores differ
- Interpret p-value